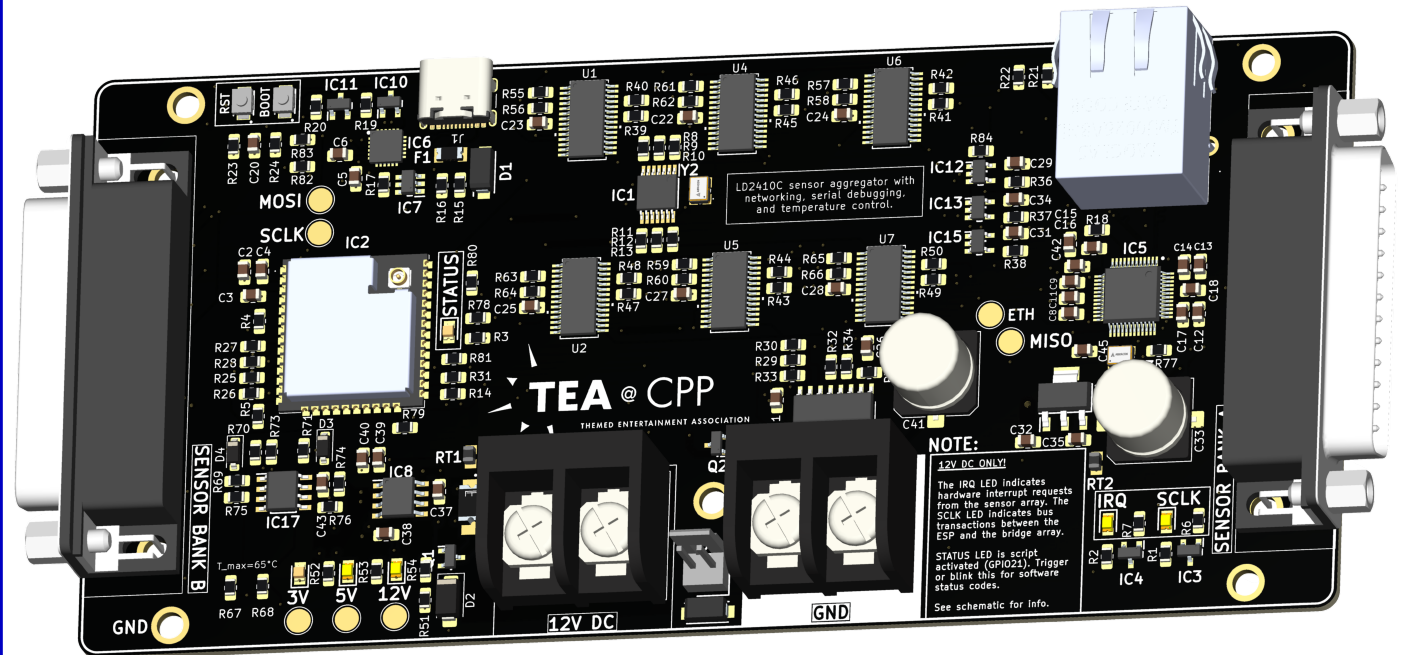


## TEA@CPP Escape Room 2026: "The Holy Grail"

An industrial-grade radar sensor aggregator board that bridges 12x LD2410C 24GHz sensors to an ESP32 via a buffered UART-to-SPI network. Features dual-stage voltage regulation, bus isolation, temperature control, and dedicated serial and networking ports.

### Major Components:

ESP32-WROOM-32UE  
SC16IS752IPW\_128  
CD74HC138QM96Q1  
SN74LVC1G125DBVT  
5PB1106PGGI8  
7447715003  
LMR33640  
AP63205WU-7  
5747842-2  
W5500  
USBL6-2SC6  
TMJ0026ABNL  
MMBT3904  
CP2102N-A02-GQFN24  
1812L150\_24MR  
LM393DR



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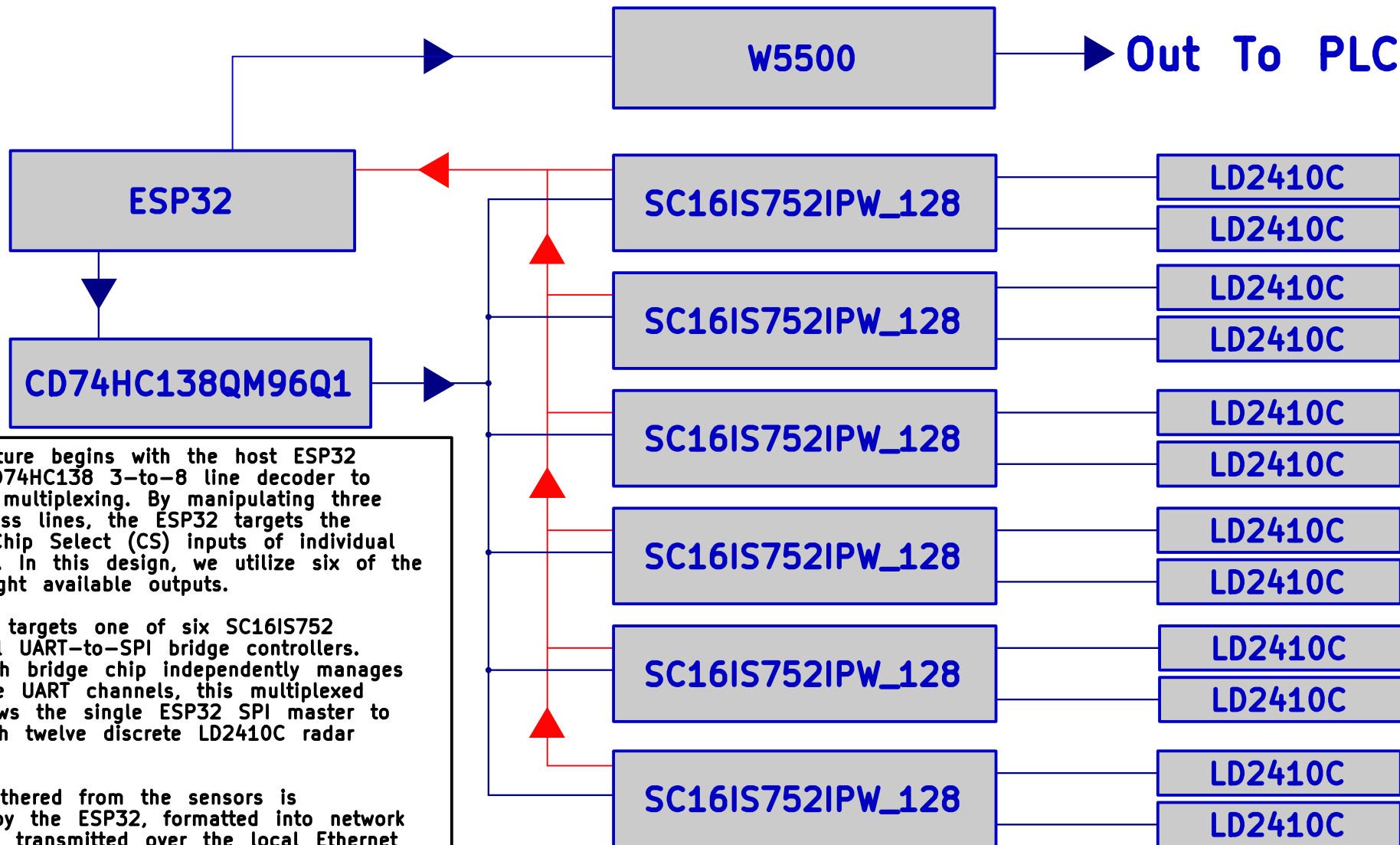


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**Title: The Holy Grail**

Size: A4 Date: 2026-04-19  
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Rev:  
Id: 1/8



The architecture begins with the host ESP32 driving a CD74HC138 3-to-8 line decoder to control bus multiplexing. By manipulating three digital address lines, the ESP32 targets the active-low Chip Select (CS) inputs of individual bridge chips. In this design, we utilize six of the decoder's eight available outputs.

Each output targets one of six SC16IS752 dual-channel UART-to-SPI bridge controllers. Because each bridge chip independently manages two hardware UART channels, this multiplexed network allows the single ESP32 SPI master to interface with twelve discrete LD2410C radar sensors.

Telemetry gathered from the sensors is aggregated by the ESP32, formatted into network packets, and transmitted over the local Ethernet show network.

**NOTE:** The original target of this board was to read 24 sensors on one board. To achieve 24 sensors, this board should be duplicated and ran in parallel. Adding 2 more SC16IS752 chips to this design increases the max theoretical capacity to 16 sensors, but requires a complete board redesign.

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**Title: Block Diagram**

Size: A4 Date: 2026-04-27  
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Rev:  
Id: 2/8

ESP32: Programming GPIO Pins

Interrupt: 4  
SCLK: 18  
MISO: 19  
MOSI: 23  
Reset: 2  
OE: 27

Ethernet Chip Select:  
5 (Active Low)  
ETH Reset: 13

Status LED: 21

Line Decoder  
A0: 25  
A1: 26  
A2: 32  
Enable: 33

Spare IO Pins: 16, 17

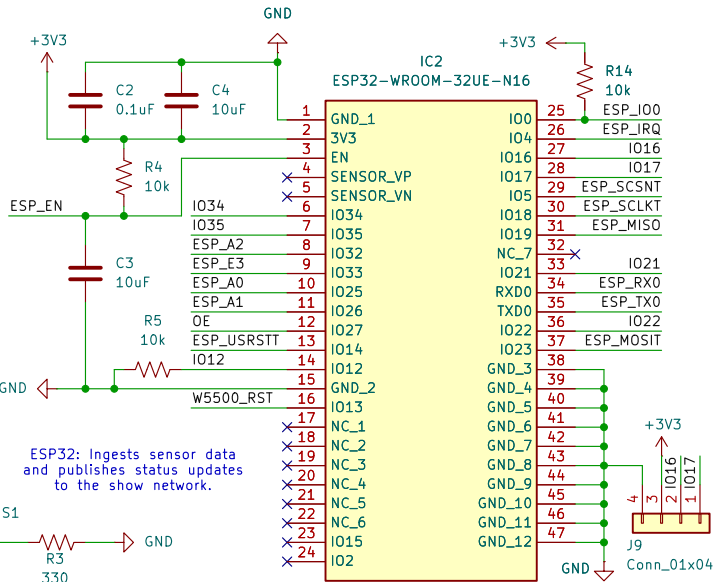
IUART Clock Frequency:  
14.7456 MHz

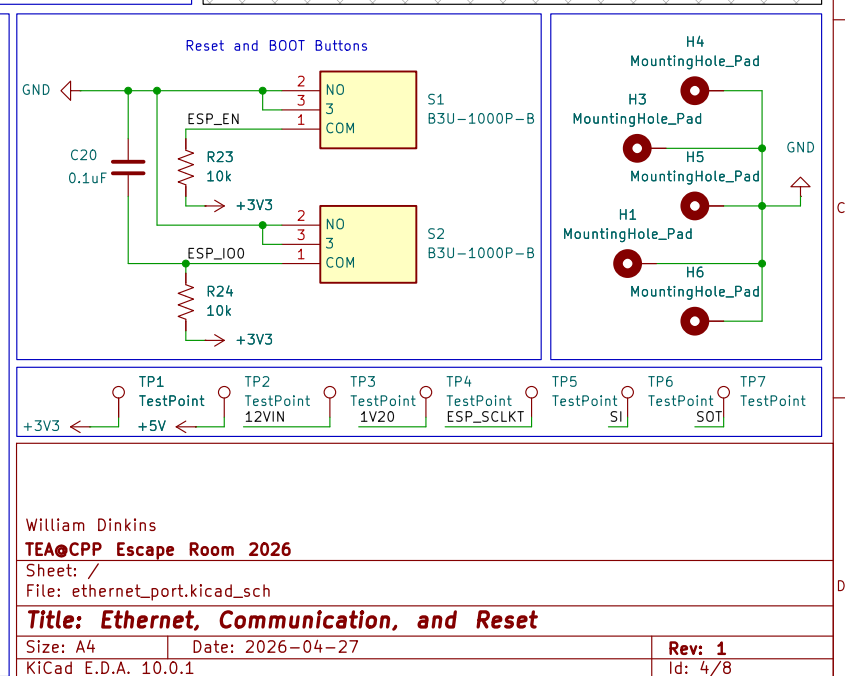
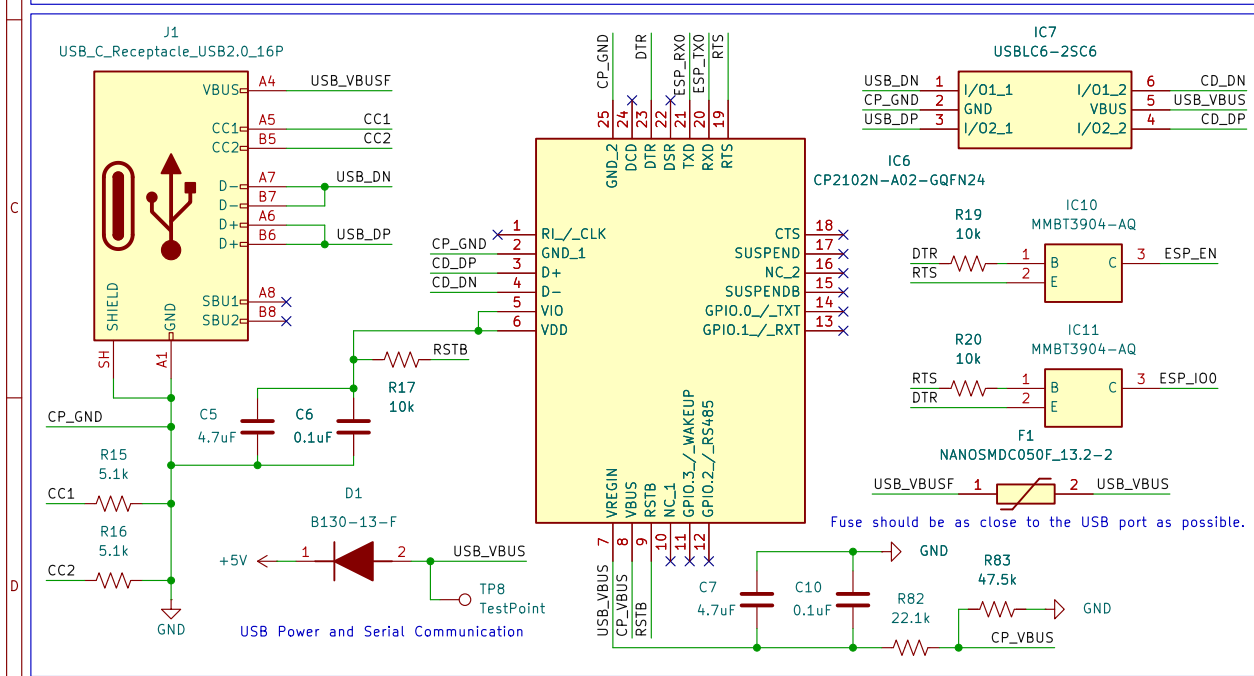
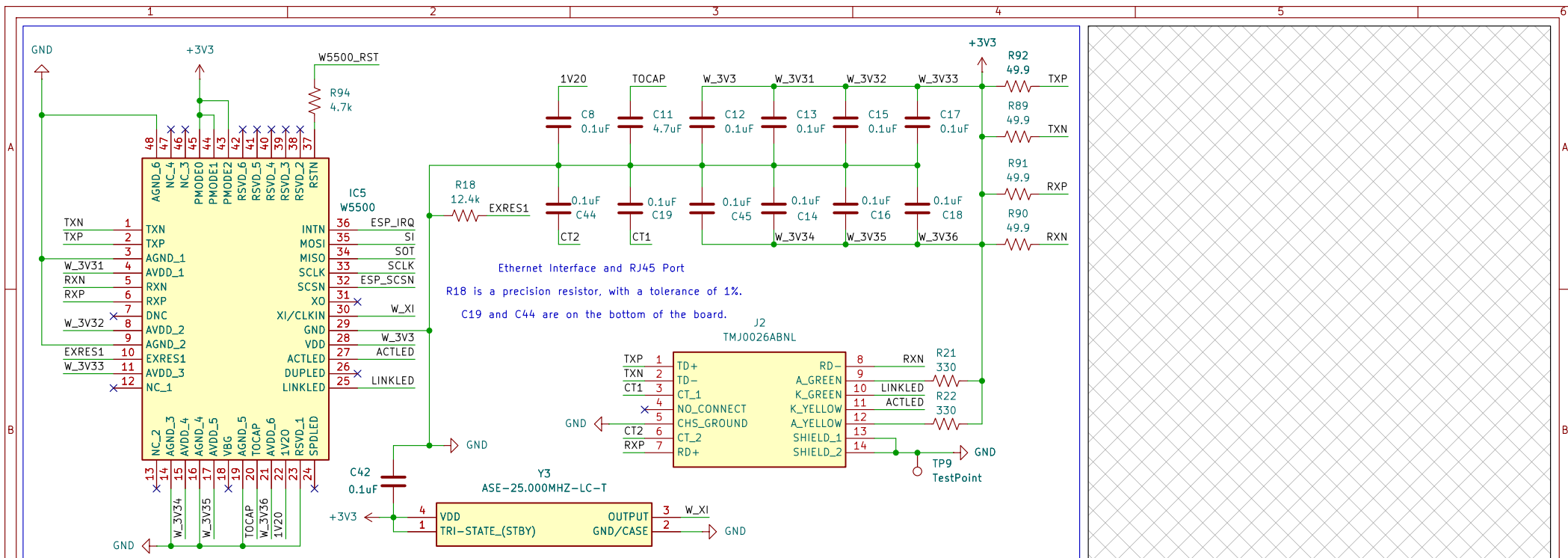
SPI speed: <20MHz  
(16 Mhz recommended)

Use Static IP Addressing

Use MQTT (Need a spare computer)

Thermal management is hardware triggered. See page 7 for more info.





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File: ethernet\_port.kicad\_sch

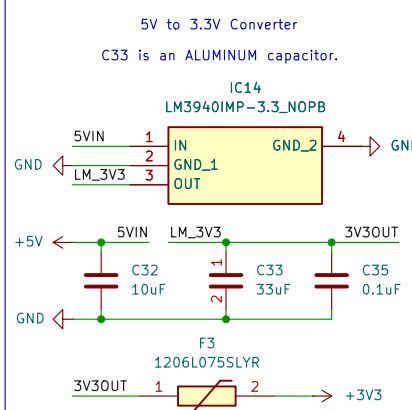
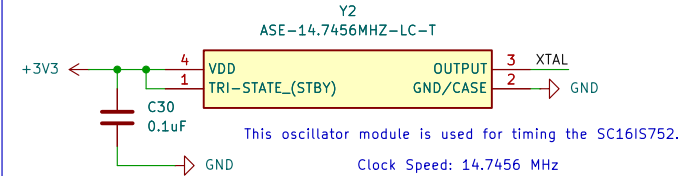
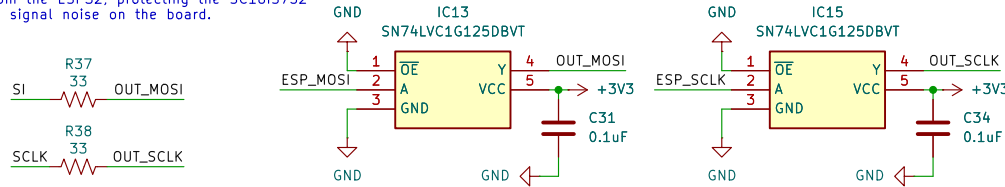
**Title: Ethernet, Communication, and Reset**

Size: A4 Date: 2026-04-27  
KiCad E.D.A. 10.0.1

Rev: 1  
Id: 4/8



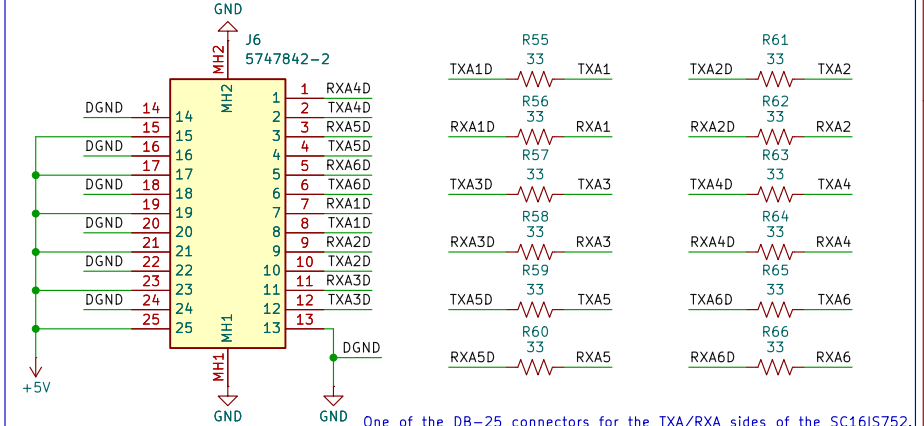
Tri-state line driver that isolates the data lines coming from the ESP32, protecting the SC16IS752 from stray signal noise on the board.



## CONNECTORS

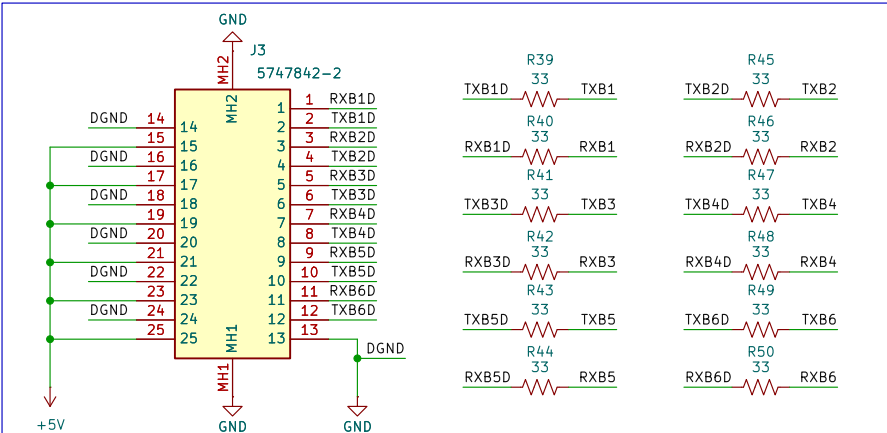
We need to connect 24 high frequency sensors to one PCB.

We'll need a wiring and connector solution that can take that kind of chaos.



Refer to page 8 for cable pinout.

This block handles sensor bank A, the RIGHT side connector. The 33 Ohm resistors are there to prevent signal reflections, increasing our signal quality.

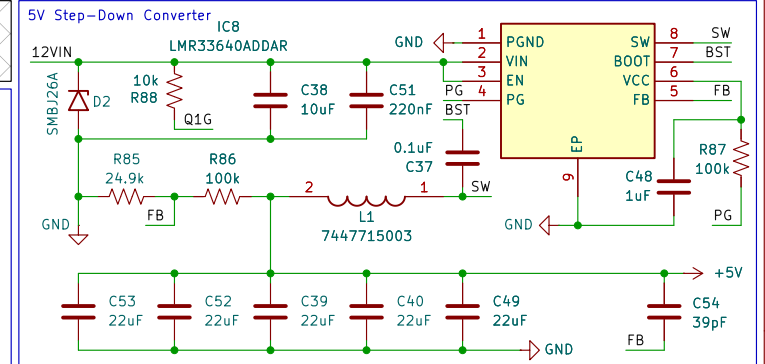
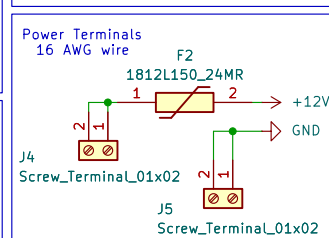
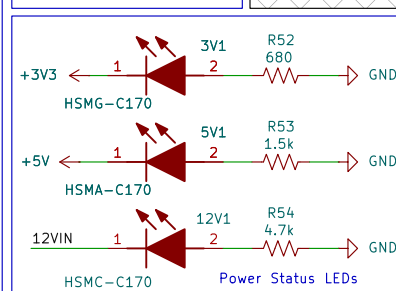
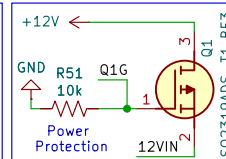


The other DB-25 connector for the TXB/RXB sides of the SC16IS752.

This block handles sensor bank B, the LEFT side connector.

The 33 Ohm resistors are there to prevent signal reflections, increasing our signal quality.

Refer to page 8 for cable pinout.



| Max Current | Wire Gauge |
|-------------|------------|
| 2A          | 18 AWG     |
| 5A          | 16 AWG     |
| 10A         | 14 AWG     |
| 15A         | 12 AWG     |

**NOTE:**  
Despite two terminals, the power terminals share the same island. This allows the terminals to "daisy-chain" to another device that requires up to 12V DC power. Be wary of voltage drop over cable length and no. of devices when doing this.  
Calculate total current draw and refer to table for appropriate wire gauges. The included terminals fit 16 AWG wire. Assume this board generates 3.5A of current.

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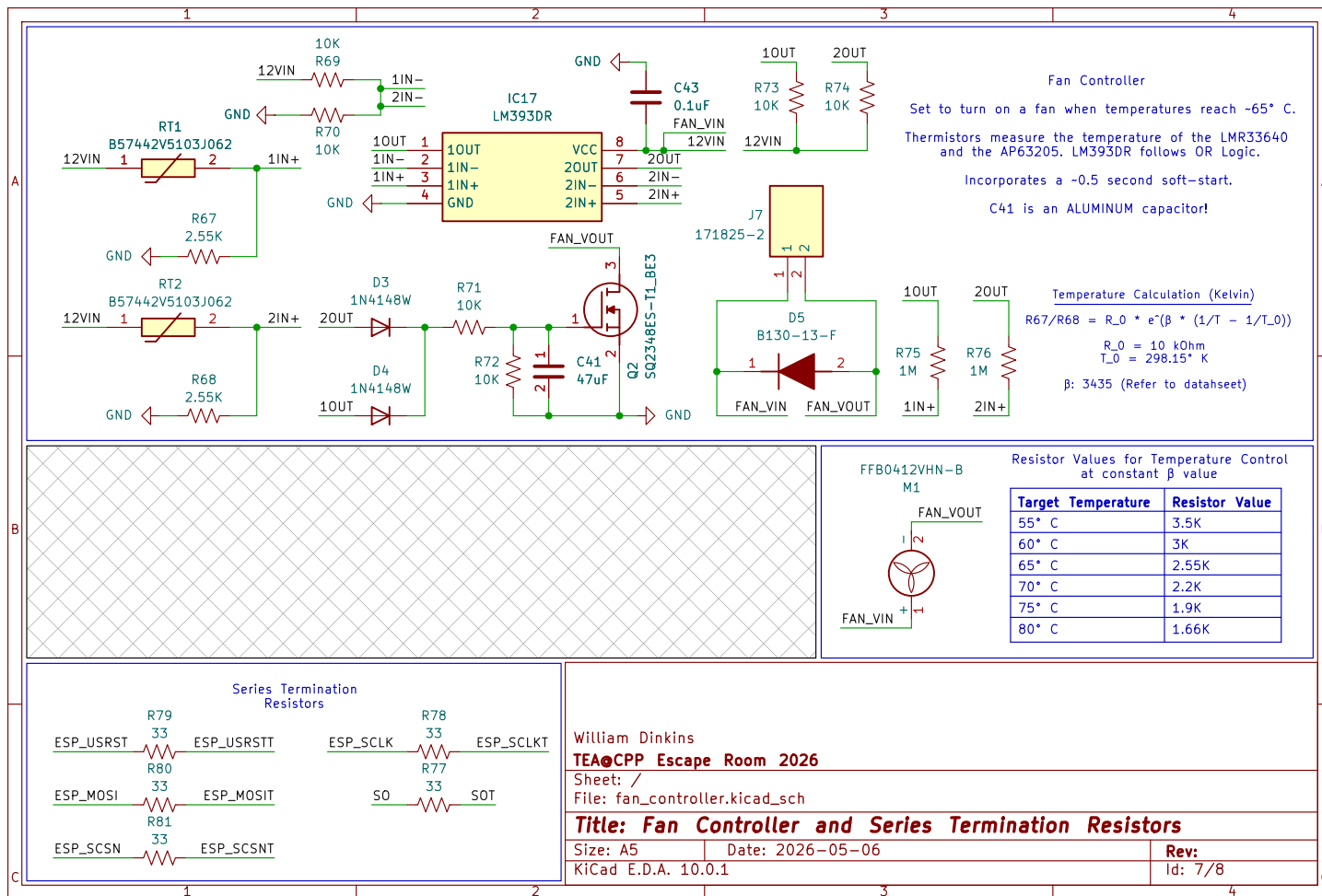
Sheet: /  
File: sc16is752\_cont..kicad\_sch

Title: Connectors, Timing, and Power Management

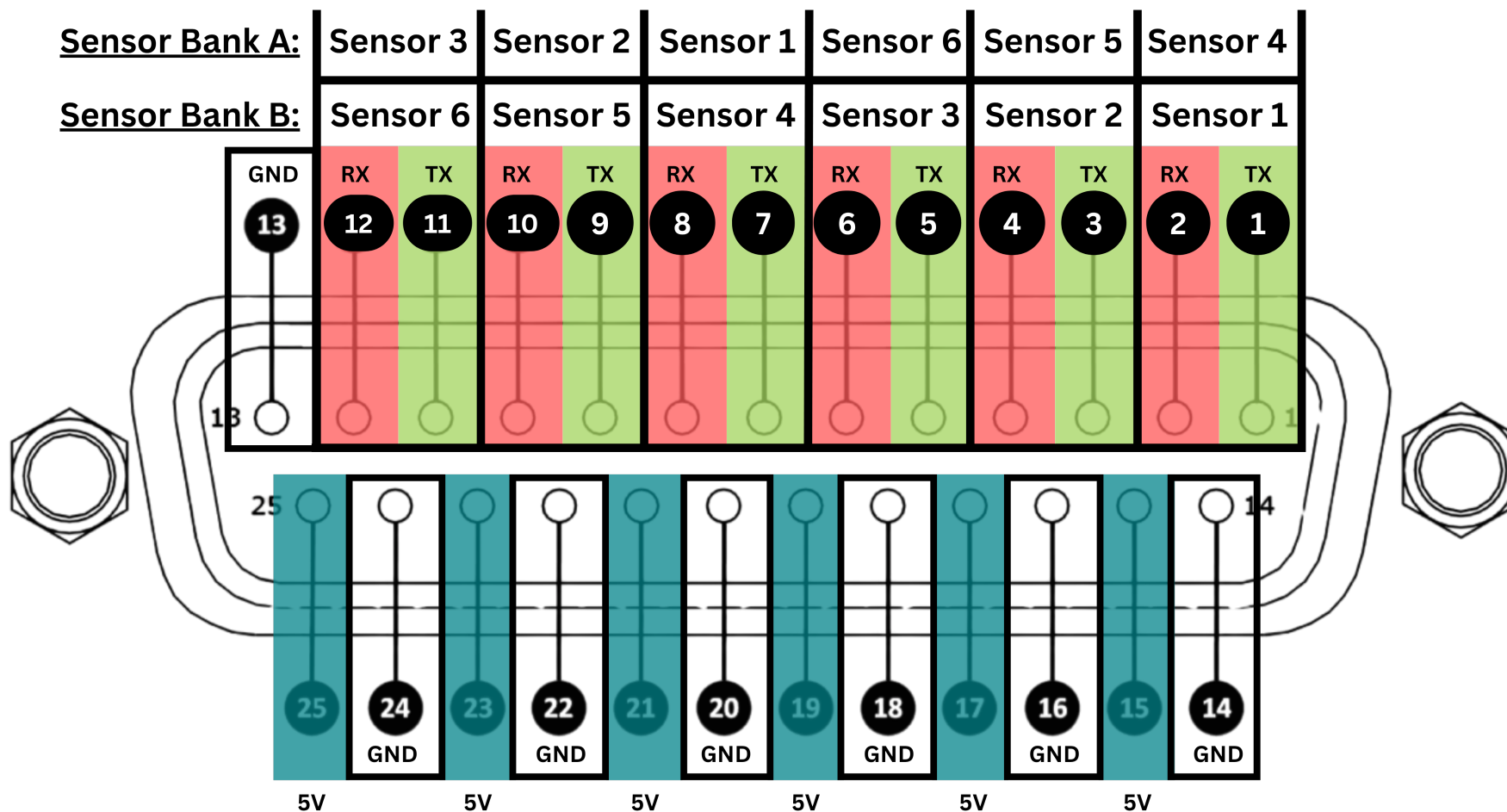
Size: A4 Date: 2026-04-26

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Rev: 1  
Id: 6/8







This is the pinout for a custom female DB25 cable. This board requires a custom cable to connect to its sensors. Use this guide when making a new cable. Cable RX goes to Board TX, same for TX to RX.

**Make sure to check all connections prior to connecting to power.**

**Incorrect wiring can cause permanent damage to sensors and/or board.**

Use a multimeter set to continuity mode to ensure connections. Have a team member verify when finished.

**Print this page out and have an installer sign here to verify connections:**

Sheet: /  
File: sensor\_pinout.kicad\_sch

**Title: DB25 Pinout**

Size: A4

Date:

KiCad E.D.A. 10.0.1

**Rev:**

Id: 8/8